

Ethan Walker

☎ 210.309.1532 | ✉ ethan.walker830@gmail.com | 🐙 github.com/SocialSalt | 🔗 [linkedin.com/in/ethanmwalker](https://www.linkedin.com/in/ethanmwalker)

EXPERIENCE

Techcyte

San Francisco, California (Remote)

Senior Software Engineer

Feb 2025 – Present

- Engineered a greenfield **DICOM Converter** microservice in **Go and C++** using **AWS Lambda, SNS, and Terraform**, eliminating slide ingestion bottlenecks and removing heap allocations to optimize memory consumption.
- Engineered queue-depth and message-age autoscaling algorithms for inference worker fleets with dynamic scale-in termination protection, maintaining consistent throughput under heavy clinical lab bursts and variable AWS instance availability.
- Implemented asynchronous workflow automation framework for enterprise clinical partners, resolving high-concurrency race conditions across background event workers.
- Architected and executed a zero-downtime cross-service migration decoupling multi-gigabyte AI and manual workflow payloads from relational database rows into an **S3 ledger architecture** with presigned URL streaming across repositories.

Software Engineer

June 2022 – Feb 2025

- Engineered an **on-premises AI inference engine** migrating workloads from AWS to local clinical edge hardware, achieving complete parity with cloud results via multi-container Docker environments.
- Collaborated on **hardware-software co-design** with scanner vendors to resolve GPU resource contention, implementing a dual-GPU pipeline that stabilized real-time scanning and AI processing throughput.
- Maintained core deep learning infrastructure and computer vision training frameworks (**PyTorch, Python**) across object detection and classification models (EfficientDet, HRNet).
- Built and delivered full-stack clinical viewer features, custom differential workflow cards, and performant GraphQL/React user interfaces in **TypeScript and Go**.

Brigham Young University

Provo, Utah

Graduate Teaching Assistant

August 2020 – April 2022

- Instructed undergraduate courses on **Machine Learning, Numerical Methods, and Optimal Control**, focusing on theoretical mathematical foundations and robust algorithmic implementations.
- Mentored students in scientific computing, software design principles, and numerical optimization techniques.

EDUCATION

MS, Mathematics

April 2022

Brigham Young University

Provo, Utah

- Coursework:** Probability Theory, Machine Learning, Analysis, Dynamical Systems
- Thesis:** Synchronization in Growing Dynamical Networks | [Published in Physica A](#)

BS, Applied and Computational Mathematics

April 2020

Brigham Young University

Provo, Utah

- Coursework:** Machine Learning, Bayesian Statistics, Data Structures, Algorithms, Optimal Control

SKILLS

Languages & Frameworks: Go, Python, TypeScript, SQL, GraphQL, React

Cloud & DevOps: AWS (S3, Lambda, SNS/SQS, EC2), Terraform, Docker, CI/CD (GitLab)

Systems, Data & AI: PyTorch, Distributed Systems, DICOM, Edge AI